

Function and description

```

bit[len]
Crypto_KDF(
    byte[] inputKey,
    byte[] salt,
    byte[] info,
    int len)

```

Returns the key of **len** bits derived from **inputKey** using the **salt** and the **info**; **len** SHALL be a multiple of 8.

Mapping (Version 1.0)

```

Crypto_KDF(inputKey, salt, info, len) :=
    bit[len] HKDF-Expand(
        PRK := HKDF-Extract(salt := salt, IKM := inputKey),
        info := info, L := (len / 8))

```

HKDF-Extract SHALL be the HKDF-Extract primitive from [RFC 5869](#) section 2.2 using **Crypto_HMAC** (i.e. HMAC-SHA256) as the auxiliary **HMAC-Hash** function.

HKDF-Expand SHALL be the HKDF-Expand primitive from [RFC 5869](#) section 2.3 using **Crypto_HMAC** (i.e. HMAC-SHA256) as the auxiliary **HMAC-Hash** function.

The primitive returns a bit array of **len** bits.

When multiple keys of the same length are generated by a single KDF call, the following shorthand notation can be used:

```

Key1 || Key2 || Key3 = Crypto_KDF
(
    inputKey = inputKeyMaterial,
    salt = [],
    info = [],
    // 3 below matches number of keys expressed in concatenated output
    len = 3 * CRYPTO_SYMMETRIC_KEY_LENGTH_BITS
)

```

This is equivalent to the following:

```

Keys = Crypto_KDF
(
    inputKey = inputKeyMaterial,

```